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## Integrable Systems — Exercise Sheet 3

*The exterior derivative, Stokes, and the symplectic form*

**Variant: Questions only**

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Conventions throughout, as in Lecture 3:  $\omega = \sum_i dq_i \wedge dp_i$  in that order; Hamilton's vector field solves  $\iota_X \omega = dH$ , with  $X$  in the *first* slot; on two dimensions  $(a dx + b dy) \wedge (c dx + d dy) = (ad - bc) dx \wedge dy$ ; and  $\iota_v(\mu \wedge \nu) = \mu(v)\nu - \nu(v)\mu$  for 1-forms  $\mu, \nu$ . On  $\mathbb{R}^3$ ,  $\text{vol} = dx \wedge dy \wedge dz$ .

**Exercise 1 (d in three dimensions).** On  $\mathbb{R}^3$  with coordinates  $(x, y, z)$  let

$$\alpha = yz dx + xz dy + xy dz, \quad \beta = -z dx + x^2 dy, \quad \gamma = x dy \wedge dz + y dz \wedge dx + z dx \wedge dy.$$

(i) Compute  $d\alpha$ . Is  $\alpha$  exact? If it is, find a primitive. (ii) Compute  $d\beta$ , and then  $d(d\beta)$  directly. Using the table in the lecture's reading on grad, curl and div, say which classical operator  $d\beta$  computes, and check the answer against the classical formula. (iii) Compute  $d\gamma$  and name the classical quantity its coefficient is. (iv) Compute  $\alpha \wedge \beta$  and  $d(\alpha \wedge \beta)$ , and check the graded Leibniz rule  $d(\alpha \wedge \beta) = d\alpha \wedge \beta - \alpha \wedge d\beta$ . Would the check catch the rule written with +?

**Exercise 2 (the circulation round a small square).** Let  $\alpha = P dx + Q dy$  with  $P = xy^2$  and  $Q = x^3$ , and let  $S_h = [a, a+h] \times [b, b+h]$ , its boundary  $\partial S_h$  traversed anticlockwise. (i) Compute  $\oint_{\partial S_h} \alpha$  exactly, edge by edge, as a polynomial in  $h$ . (ii) Show that  $h^{-2} \oint_{\partial S_h} \alpha \rightarrow (\partial_x Q - \partial_y P)(a, b)$  as  $h \rightarrow 0$ , and that this limit is  $d\alpha(\partial_x, \partial_y)$  at  $(a, b)$ : the lecture's reading of  $d\alpha$  as circulation per unit area. (iii) Compute  $\int_{S_h} d\alpha$  and compare it with (i). Which theorem have you verified? (iv) For a general smooth  $\alpha$ , pair opposite edges and expand to first order to show  $\oint_{\partial S_h} \alpha = h^2(\partial_x Q - \partial_y P)(a, b) + O(h^3)$ , although each edge alone is of order  $h$ .

**Exercise 3 (what the angle form sees).** Let  $\alpha = (x dy - y dx)/(x^2 + y^2)$  be the angle form on  $\mathbb{R}^2 \setminus \{0\}$ , and let  $t \in [0, 2\pi]$  throughout. (i) Compute  $\oint \alpha$  over  $\gamma_a(t) = (3 + \cos t, \sin t)$ ,  $\gamma_b(t) = (\cos 2t, \sin 2t)$ ,  $\gamma_c(t) = (2 \cos t, -\sin t)$ , and  $\gamma_m(t) = (\cos mt, \sin mt)$  for an integer  $m \geq 1$ . For  $\gamma_a$ , find a primitive of  $\alpha$  on the half-plane  $x > 0$  instead of integrating. For  $\gamma_c$  you may use  $\int_0^{2\pi} dt/(A^2 \cos^2 t + B^2 \sin^2 t) = 2\pi/(AB)$  for  $A, B > 0$ . (ii) Obtain the answer for  $\gamma_c$  again without integrating: write down a deformation from the unit circle traversed clockwise to  $\gamma_c$ , check that it never meets the origin, and apply the lecture's corollary on deformation invariance. (iii) The circle  $r = 2$  traversed anticlockwise and the circle  $r = 1$  traversed clockwise together make up the boundary of the annulus  $A = \{1 \leq r \leq 2\}$ , with its induced orientation. Compute both integrals and check Stokes' theorem on  $A$ . Why does the deformation corollary *not* say that the two integrals are equal? (iv) Now let  $\gamma$  be *any* smooth closed curve in  $\mathbb{R}^2 \setminus \{0\}$ , written in polar form  $\gamma(t) = r(t)(\cos \theta(t), \sin \theta(t))$  with smooth  $r > 0$  and  $\theta$  on  $[0, 2\pi]$ . (Every such curve can be written so; take that as given.) Closedness forces  $\theta(2\pi) - \theta(0) = 2\pi m$  for an integer  $m$ , the *winding number* of  $\gamma$ . Show that  $\oint_\gamma \alpha = 2\pi m$ , however the loop wanders or crosses itself, and that  $F(t, s) = ((1-s)r(t)+s)(\cos \theta(t), \sin \theta(t))$ ,  $s \in [0, 1]$ , is a deformation from  $\gamma$  to a loop on the unit circle that never meets the origin.

**Exercise 4 (pulling back a 2-form).** (i) Let  $F(u, v) = (x(u, v), y(u, v))$  be smooth. Show that  $F^*(dx \wedge dy) = \det(\partial(x, y)/\partial(u, v)) du \wedge dv$ , and deduce  $dx \wedge dy = r dr \wedge d\theta$  in polar coordinates. (ii) Let  $\nu > 0$ , and on  $\mathbb{R}^2$  minus the origin introduce  $(\varphi, I)$ ,  $I > 0$ , by

$$q = \sqrt{2I/\nu} \sin \varphi, \quad p = \sqrt{2I\nu} \cos \varphi.$$

Show that  $dq \wedge dp = d\varphi \wedge dI$ , and that the choice  $q = \sqrt{2I/\nu} \cos \varphi$ ,  $p = \sqrt{2I\nu} \sin \varphi$  gives  $-d\varphi \wedge dI$  instead. (iii) With the first choice and  $H = \frac{1}{2}(p^2 + \nu^2 q^2)$ , express  $H$  in  $(\varphi, I)$  and solve  $\iota_X \omega = dH$  there, using  $\omega = d\varphi \wedge dI$  and the lecture's sign test with  $\varphi$  in the place of  $q$ . Check at  $\varphi = 0$  that increasing  $\varphi$  is the direction of the motion. (iv) Compute  $\oint p dq$  round the curve  $I = \text{const}$ , traversed with  $\varphi$  increasing, and compare it with the area of the region the curve encloses.

**Exercise 5** ( *$p dq$  on the pendulum's cylinder*). The pendulum  $H = \frac{1}{2}p^2 - \cos q$  has phase space the cylinder  $S^1 \times \mathbb{R}$ , with  $q$  an angle defined modulo  $2\pi$ . (i)  $q$  is not a function on the cylinder. Explain why  $dq$  and  $p dq$  are nevertheless 1-forms there, and compute  $d(p dq)$ . Is  $p dq$  closed on the cylinder? (ii) For  $E > 1$  the level  $H = E$  has a component  $\gamma_E$  on which  $p > 0$ : the pendulum rotates. Write  $\oint_{\gamma_E} p dq$ , traversed in the direction of the motion, as an integral over  $q \in [0, 2\pi]$ , and show that it lies strictly between  $2\pi\sqrt{2(E-1)}$  and  $2\pi\sqrt{2E}$ . Deduce that the pullback of  $p dq$  to the circle  $\gamma_E$  is closed but not exact. (iii) Now let  $H = p$  on the same cylinder. Show that every level set  $\{p = c\}$  is an invariant circle, and compute  $\oint p dq$  round it in the direction of the motion. What does the level  $c = 0$  say about the claim that the loop integral of  $p dq$  round an invariant circle cannot vanish?

**Exercise 6** (*odd dimensions are degenerate*). (i) Show that an antisymmetric  $m \times m$  real matrix is singular when  $m$  is odd. Deduce that a manifold carrying a non-degenerate 2-form has even dimension. (ii) On  $\mathbb{R}^3$  find every vector  $v$  with  $\iota_v \beta = 0$ , where  $\beta = dx \wedge dy + dy \wedge dz + dz \wedge dx$ . (iii) On  $\mathbb{R}^4$  with coordinates  $(q_1, q_2, p_1, p_2)$  find every  $v$  with  $\iota_v \beta' = 0$ , where  $\beta' = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ . Is  $\beta'$  closed? Is it symplectic?

**Exercise 7** ★. *A magnetic symplectic form*. On  $\mathbb{R}^4$  with coordinates  $(q_1, q_2, p_1, p_2)$  and a real constant  $B$ , let

$$\omega_B = dq_1 \wedge dp_1 + dq_2 \wedge dp_2 + B dq_1 \wedge dq_2.$$

(i) Show that  $\omega_B$  is symplectic. (ii) For  $H = \frac{1}{2}(p_1^2 + p_2^2)$ , solve  $\iota_X \omega_B = dH$  for  $X$ , and solve the equations of motion it gives. Describe the orbits in the  $(q_1, q_2)$ -plane, and show that  $H$  is conserved — first by computing, then from the antisymmetry of  $\omega_B$  alone. (iii) Let  $B = b(q_1, q_2)$  be a smooth function instead. Is  $\omega_B$  still symplectic? What if  $b$  depends on  $p_1$  as well?